A corrected modulo-sixteen classification for A396844

Dev Goyal

24 October 2026

AI disclosure and review status. ChatGPT (OpenAI) generated the mathematical arguments, formal proof code, and this manuscript from Dev Goyal's prompts. Dev Goyal is the sole human author and initiated the investigation. This consolidated presentation is AI-generated; independent human review of this presentation is pending. Source conjectures and prior results retain the credits given below.

Abstract

We give a separate complete manuscript for A396844, including the parameter-family proof on which its conclusions depend. For the unique normalized solution of $A_m(xA_m - mxA_m^2) = x^2$, we prove parity, exact residues modulo m^2, binary-digit divisibility for both the series and its compositional inverse, and the minimum prime-exponent criterion. The entry under consideration is $m = 4$.

1 The entry and the claims

Let $a_n(m) = [x^n]A_m(x)$, where

$$A_m(xA_m(x) - mxA_m(x)^2) = x^2, \qquad A_m = x + O(x^2). \tag{1}$$

The sequence A396844 is the case $m = 4$; the constant coefficient is zero, and the OEIS sequence is indexed from one. The defining series and source questions are due to Paul D. Hanna [4]. We write $s_2(n)$ for the number of ones in the binary expansion of n. The general results below give, in particular,

$$m^{s_2(n-1)} \mid a_n(m) \quad (n \geq 1), \qquad a_{2^j+1}(m) \equiv m(-1)^j \pmod{m^2}.$$

All other coefficients of degree at least two vanish modulo m^2. The binary bound also holds for the coefficients of the inverse series.

The corrected residue theorem is, for $n \geq 2$,

$$a_n(4) \equiv \begin{cases} 4 \pmod{16}, & n = 4^k + 1,\ k \geq 0, \\ 12 \pmod{16}, & n = 2 \cdot 4^k + 1,\ k \geq 0, \\ 0 \pmod{16}, & \text{otherwise.} \end{cases}$$

The first exceptional class is essential. The earlier source assertion specified residue twelve in the second class and divisibility by sixteen otherwise for $n > 8$. At $n = 17 = 4^2 + 1$ the correct residue is four, so that other clause is false. We prove the corrected theorem, not the false original clause. All coefficients after the first are even.

Provenance. These arguments are adapted from the original A396843 repository supplied by Dev Goyal. Its historical declaration of human verification concerns that earlier report. It is not a declaration that this new editorial consolidation has received independent human review. The parameter-family results are retained mathematical work, not new problems introduced by this final packaging.

2 Formal power series and notation

2.1 Operations and coefficient comparison

A formal power series is an expression $U(x) = \sum_{n\geq 0} u_n x^n$ whose coefficients are considered algebraically. We do not substitute numerical values of x, and no convergence assumption is needed. Equality means equality of every coefficient. We write $[x^n]U = u_n$. Addition is coefficientwise, and multiplication is defined by

$$[x^n](UV) = \sum_{i=0}^{n} u_i v_{n-i}.$$

Thus each coefficient of a product is a finite sum. An empty sum is zero.

If $V(0) = 0$, composition is defined by

$$U(V(x)) = \sum_{n\geq 0} u_n V(x)^n.$$

The power $V(x)^n$ has no terms of degree below n, so each coefficient again receives only finitely many contributions. For example, if $U(x) = x + x^2$ and $V(x) = x^2$, then $U(V(x)) = x^2 + x^4$. Composition of series with zero constant coefficient is associative: one can verify this in each degree by the usual rules for polynomials, since only finitely many terms contribute.

The notation $O(x^d)$ denotes an unspecified series whose coefficients below degree d vanish. In particular, $U = x + O(x^2)$ means that U has constant coefficient zero and coefficient of x equal to one. We call this *normalized*. Products and compositions satisfy

$$O(x^a)O(x^b) = O(x^{a+b}), \qquad U(V(x)) = O(x^{ab}) \quad \text{if } U = O(x^a),\ V = O(x^b),\ b \geq 1.$$

These statements specify vanishing of coefficients, not estimates as a numerical variable tends to zero.

We occasionally allow coefficients in a commutative ring R with identity: a coefficient system in which addition, subtraction, and multiplication obey their usual algebraic laws, with a multiplicative identity 1. The examples used here are the integers $\mathbb{Z}$, residues $\mathbb{Z}/q\mathbb{Z}$, and integer polynomials $\mathbb{Z}[t]$. The notation $R[[x]]$ denotes formal power series with coefficients in R.

For integer series, $U \equiv V \pmod{q}$ means that every coefficient of $U - V$ is divisible by q. Reducing coefficients modulo q commutes with addition, multiplication, and every composition used here: each coefficient is obtained by a finite sequence of integer operations. In particular, $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ has the two elements 0 and 1, and addition equals subtraction in this ring.

If $U(0) = 0$, the series $1 + U$ has a multiplicative inverse, given formally by

$$(1 + U)^{-1} = \sum_{k\geq 0} (-U)^k.$$

The expression is well defined coefficientwise, and multiplication makes successive terms cancel. This justifies fractions such as $x/(1 + x)$.

2.2 Compositional inverses

A compositional inverse reverses substitution rather than multiplication. We will need both identities in the following elementary fact.

Lemma 2.1. *For any commutative ring R with identity and any normalized series $A \in R[[x]]$, there is a unique normalized series $B \in R[[x]]$ such that*

$$A(B(x)) = B(A(x)) = x.$$

Proof. Write $A = x + \sum_{n\geq 2} a_n x^n$ and seek $B = x + \sum_{n\geq 2} b_n x^n$ with $A(B) = x$. In degree $n \geq 2$, the linear term B contributes b_n. In every higher power B^k, with $k \geq 2$, a contribution to degree n uses only coefficients of B of index at most $n-k+1 < n$. Consequently, the coefficient equation in degree n determines b_n uniquely from earlier coefficients, using only addition, subtraction, and multiplication. This constructs the unique normalized B with $A \circ B = x$.

Apply the same construction to B to obtain a normalized C with $B \circ C = x$. Associativity gives

$$A = A \circ (B \circ C) = (A \circ B) \circ C = C.$$

Hence $B \circ A = B \circ C = x$ as well. Uniqueness follows from the construction of the right inverse. □

3 Existence and uniqueness of the defining series

The following coefficient comparison is the reason modular solutions can be identified by direct substitution.

Lemma 3.1. *Let R be a commutative ring with identity and let $m \in R$. There is exactly one normalized series $A \in R[[x]]$ satisfying*

$$A(xA - mxA^2) = x^2.$$

Proof. For a normalized series A, set $T_A = xA - mxA^2$ and $\Psi_m(A) = A(T_A)$. We first show how a change in one coefficient affects the defining equation:

$$B - A = cx^r + O(x^{r+1}), \quad r \geq 2 \quad \Longrightarrow \quad \Psi_m(B) - \Psi_m(A) = cx^{r+1} + O(x^{r+2}). \tag{2}$$

Indeed,

$$T_B - T_A = x(B - A)(1 - m(A + B)) = cx^{r+1} + O(x^{r+2}).$$

Both T_A and T_B start with x^2. If $A = x + \sum_{k\geq 2} a_k x^k$, then

$$A(T_B) - A(T_A) = (T_B - T_A) + \sum_{k\geq 2} a_k (T_B^k - T_A^k).$$

For $k \geq 2$, factor $T_B^k - T_A^k$ as $T_B - T_A$ times a sum of products of $k-1$ inner arguments. Its degree is therefore at least $r + 1 + 2(k-1) \geq r + 3$. It follows that

$$A(T_B) - A(T_A) = (T_B - T_A) + O(x^{r+3}).$$

The other difference, $(B - A)(T_B)$, is $O(x^{2r})$, hence $O(x^{r+2})$. Adding the two differences proves (2).

For uniqueness, if two solutions were different, take the smallest index r at which their coefficients differ. In (2), c is then nonzero, so their images under Ψ_m differ in degree $r+1$. They cannot both have image x^2.

For existence, start with $A = x$, for which $\Psi_m(A) = x^2 - mx^3$. Suppose the residual $\Psi_m(A) - x^2$ has already been made zero through degree r. Let c be its coefficient of x^{r+1}. Replacing A by $A - cx^r$ removes that coefficient without changing any earlier one, by (2). Continue for $r = 2, 3, \ldots$. Each coefficient of A eventually stops changing. Each coefficient of its defining equation depends on only finitely many coefficients of A, so the resulting series satisfies the equation in every degree.

More precisely, for $d \geq 2$, the coefficient $[x^d]\Psi_m(A)$ depends only on $a_0, \ldots, a_{d-1}$. The inner series T_A vanishes below degree two, so only outer indices $k \leq d/2$ contribute. Each coefficient of T_A through degree d depends only on coefficients of A through degree $d-1$. Thus later corrections to A cannot change a coefficient of the defining equation that has already been fixed. □

The proof uses no division by a coefficient. In particular it applies in residue rings even when the modulus is composite. Applying the lemma over $\mathbb{Z}$ defines A_m for every integer m. Its first terms are

$$A_m(x) = x + mx^2 + (2m^2 - m)x^3 + (5m^3 - 2m^2)x^4 + O(x^5). \tag{3}$$

For example, the coefficient of x^3 in the defining composition is $a_2(m) - m$, so $a_2(m) = m$.

Applying the same construction with an indeterminate parameter t over $\mathbb{Z}[t]$ shows that each $a_n(t)$ is an integer polynomial. Substitution $t = m$ then gives $a_n(m)$, by uniqueness. This observation will also allow a polynomial version of the divisibility theorem.

4 Parity

Theorem 4.1. *If m is odd, every $a_n(m)$ with $n \geq 1$ is odd. If m is even, then $a_1(m) = 1$ and every $a_n(m)$ with $n \geq 2$ is even.*

Proof. Suppose first that m is odd. Write P for the series obtained by reducing each coefficient of A_m modulo two. Equation (1) becomes

$$P(xP(x) + xP(x)^2) = x^2 \quad \text{in } \mathbb{F}_2[[x]], \qquad P = x + O(x^2). \tag{4}$$

The normalized solution is unique by Lemma 3.1. Consider the explicit series

$$H(x) = \frac{x}{1+x} = \sum_{n \geq 1} x^n \quad \text{in } \mathbb{F}_2[[x]].$$

The last equality is the geometric series identity from Section 2, with all signs equal modulo two. Direct calculation gives

$$xH(x) + xH(x)^2 = \frac{x^2(1+x) + x^3}{(1+x)^2} = \frac{x^2}{(1+x)^2} =: T(x).$$

Therefore

$$H(T(x)) = \frac{T(x)}{1+T(x)} = \frac{x^2}{(1+x)^2 + x^2} = x^2,$$

since $(1+x)^2 + x^2 = 1$ in $\mathbb{F}_2[[x]]$. Thus H satisfies (4), and uniqueness gives $P = H$. Every nonconstant coefficient of H is one, so every $a_n(m)$ with $n \geq 1$ is odd.

If m is even, the reduced equation is $P(xP(x)) = x^2$. The series $P = x$ satisfies it and is the unique normalized solution by Lemma 3.1. Its coefficients after the first are zero, which proves the assertion for even m. □

Taking $m = 3$ proves the coefficient-parity conjecture in OEIS A396843.

5 An exact formula modulo m^2

The parity proof identifies a solution after reduction modulo two. We now use the same uniqueness principle modulo the parameter m, and retain one further level of information modulo m^2.

Theorem 5.1. *For every integer $m \geq 2$,*

$$A_m(x) \equiv x + m \sum_{j \geq 0} (-1)^j x^{2^j+1} \pmod{m^2}. \tag{5}$$

Equivalently, for $n \geq 2$,

$$a_n(m) \equiv \begin{cases} m(-1)^j \pmod{m^2}, & n = 2^j + 1 \text{ for some } j \geq 0, \\ 0 \pmod{m^2}, & \text{otherwise.} \end{cases} \tag{6}$$

Proof. Reduce (1) modulo m. The resulting equation is $P(xP) = x^2$, whose unique normalized solution is $P = x$. Hence $A_m \equiv x \pmod{m}$, and we can write

$$A_m = x + mC, \qquad C \in x^2\mathbb{Z}[[x]].$$

The notation $C \in x^2\mathbb{Z}[[x]]$ means that C has integer coefficients and no terms of degree below two. Working modulo m^2 gives

$$T := xA_m - mxA_m^2 \equiv x^2 + m(xC - x^3).$$

Since $T \equiv x^2 \pmod{m}$, we also have $C(T) \equiv C(x^2) \pmod{m}$. Consequently,

$$A_m(T) = T + mC(T) \equiv x^2 + m\big(xC - x^3 + C(x^2)\big) \pmod{m^2}.$$

Equation (1) implies

$$xC(x) + C(x^2) \equiv x^3 \pmod{m}. \tag{7}$$

Here cancellation of m is ordinary integer divisibility: if m^2 divides md, then m divides d. It is applied separately to each coefficient.

Set

$$S(x) = \sum_{j\geq 0} (-1)^j x^{2^j+1} = x^2 - x^3 + x^5 - x^9 + x^{17} - \cdots.$$

The exponents increase, so this is a well-defined formal series. Its successive terms give

$$xS(x) = x^3 - x^4 + x^6 - x^{10} + \cdots, \qquad S(x^2) = x^4 - x^6 + x^{10} - \cdots.$$

Thus $xS + S(x^2) = x^3$ exactly.

The solution of (7) with no terms below degree two is unique over $\mathbb{Z}/m\mathbb{Z}$. Indeed, if a nonzero difference D began in degree $d \geq 2$, then xD would begin in degree $d+1$, while $D(x^2)$ would begin in degree $2d > d+1$. Its first term could not cancel in $xD + D(x^2)$. Therefore $C \equiv S \pmod{m}$, and $A_m = x + mC$ yields (5). □

6 Divisibility from binary digits

6.1 The binary digit inequality

Every nonnegative integer r has a unique binary expansion as a sum of distinct powers of two. Repeated division by two gives the binary digits as successive remainders, which also makes the expansion unique. Define $s_2(r)$ to be the number of powers in that sum, and set $s_2(0) = 0$. For example,

$$13 = 8 + 4 + 1 = (1101)_2, \qquad s_2(13) = 3.$$

Two properties will be used:

$$s_2(2r) = s_2(r), \qquad s_2(r_1 + \cdots + r_k) \leq \sum_{i=1}^{k} s_2(r_i) \quad (r_i \geq 0). \tag{8}$$

Multiplication by two shifts every binary digit one place, proving the first identity. For the inequality, expand each summand into powers of two. Whenever a power occurs twice, replace the two copies by one copy of the next power. Each replacement preserves the value and decreases the number of summands. The process terminates at the binary expansion of the total, which proves the inequality.

6.2 The divisibility theorem

Let $B_m = x + \sum_{n\geq 2} b_n(m)x^n$ be the compositional inverse of A_m, supplied by Lemma 2.1, and set $b_1(m) = 1$.

Theorem 6.1. *For every integer $m \geq 2$ and every $n \geq 1$,*

$$m^{s_2(n-1)} \mid a_n(m) \qquad \text{and} \qquad m^{s_2(n-1)} \mid b_n(m). \tag{9}$$

Proof. Fix $m \geq 2$. Remove the initial factor of x from the two inverse series:

$$A_m(x) = xF(x), \qquad B_m(x) = xG(x),$$

$$F(x) = \sum_{r\geq 0} f_r x^r, \qquad G(x) = \sum_{r\geq 0} g_r x^r, \qquad f_0 = g_0 = 1.$$

Then $f_r = a_{r+1}(m)$ and $g_r = b_{r+1}(m)$, so the shifted index r is the integer whose binary digits occur in the desired bound.

Apply B_m to (1). Using $B_m(A_m(x)) = x$ gives

$$xA_m - mxA_m^2 = B_m(x^2).$$

Hanna's entries record this identity for $m = 1, 2, 3, 4, 5$ [1, 2, 3, 4, 5]. We use it here for general m. Substitution of $A_m = xF$ and $B_m = xG$, followed by cancellation of the common factor x^2, yields

$$F(x) = G(x^2) + mxF(x)^2. \tag{10}$$

Cancellation here simply shifts every coefficient two places. For $r \geq 1$, comparison of coefficients gives

$$f_r = \begin{cases} g_{r/2}, & r \text{ even}, \\ 0, & r \text{ odd} \end{cases} + m \sum_{i+j=r-1} f_i f_j, \tag{11}$$

where i, j are nonnegative integers.

The other inverse identity, $A_m(B_m(x)) = x$, gives

$$1 = \sum_{j\geq 0} f_j x^j G(x)^{j+1}.$$

In degree $r \geq 1$, the $j = 0$ term contributes g_r and the $j = r$ term contributes f_r, because $G(0) = 1$. Thus

$$g_r = -f_r - \sum_{j=1}^{r-1} f_j [x^{r-j}] G(x)^{j+1}. \tag{12}$$

We prove by induction on r that

$$m^{s_2(r)} \mid f_r \quad \text{and} \quad m^{s_2(r)} \mid g_r. \tag{13}$$

The case $r = 0$ says that 1 divides $f_0 = g_0 = 1$. Suppose the assertions hold at every smaller index.

First consider f_r in (11). If r is even, then $r/2 < r$ and $s_2(r/2) = s_2(r)$, so the term $g_{r/2}$ has the required divisor. Each product mf_if_j uses $i, j < r$ and is divisible by $m^{1+s_2(i)+s_2(j)}$. The digit inequality (8) gives

$$1 + s_2(i) + s_2(j) \geq s_2(1 + i + j) = s_2(r).$$

Every term in (11) is therefore divisible by $m^{s_2(r)}$, and so is their sum f_r.

Now consider g_r in (12). The first term $-f_r$ has just been handled. For $1 \le j < r$, expansion of G^{j+1} expresses each remaining summand as a sum of products

$$f_j g_{i_1} \cdots g_{i_{j+1}}, \qquad j + i_1 + \cdots + i_{j+1} = r, \qquad i_\ell \ge 0.$$

Each g index is at most $r - j < r$. By the induction hypothesis the product is divisible by

$$m^{s_2(j)+s_2(i_1)+\cdots+s_2(i_{j+1})},$$

and its exponent is at least $s_2(r)$ by (8). Taking sums and negatives preserves divisibility, proving the assertion for g_r. This completes the induction: at each stage we establish f_r first and then g_r.

Finally, replace r by $n-1$ in (13). □

Remark 6.2. The same argument works before the parameter is assigned an integer value. Let t be an indeterminate and construct A_t and B_t over $\mathbb{Z}[t]$. All coefficient recurrences above use only sums, products, and negatives. The induction therefore proves that both $a_n(t)$ and $b_n(t)$ are multiples of $t^{s_2(n-1)}$ in $\mathbb{Z}[t]$. In other words, neither polynomial has a nonzero term of degree below $s_2(n-1)$ in the parameter t.

For illustration, the original parameter $m = 3$ gives the following bounds.

n	Binary expansion of $n-1$	$s_2(n-1)$	Guaranteed divisor of $a_n(3)$
2	1	1	3
8	111	3	27
9	1000	1	3
32	11111	5	243
64	111111	6	729

In general, $2^k - 1$ has k binary ones, so $m^k \mid a_{2^k}(m)$ for every $k \ge 0$. The table also shows why the guaranteed divisor need not grow with the index: a power of two has only one binary one.

6.3 Exactly where the smallest prime exponent occurs

For a prime p and a nonzero integer q, write $v_p(q)$ for the number of factors of p in q. Equivalently, it is the largest integer $e \ge 0$ such that $p^e \mid q$. For example, $v_3(45) = 2$. Set $v_p(0) = +\infty$. This notation lets us distinguish a lower bound from an exact exponent.

Corollary 6.3. *Let $m \ge 2$, let p be a prime dividing m, and put $e = v_p(m)$. Then*

$$v_p(a_n(m)) \ge e\, s_2(n-1) \qquad (n \ge 1).$$

For $n \ge 2$, equality with the smallest possible value e is characterized by

$$v_p(a_n(m)) = e \quad \Longleftrightarrow \quad n = 2^j + 1 \text{ for some } j \ge 0. \tag{14}$$

Proof. The lower bound follows immediately from Theorem 6.1. At $n = 2^j + 1$, Theorem 5.1 gives

$$a_n(m) = m\big((-1)^j + mq\big)$$

for some integer q. The factor in parentheses is 1 or -1 modulo p, so it is not divisible by p. Thus $a_n(m)$ has exactly e factors of p. At any other index $n \ge 2$, the positive integer $n-1$ is not a power of two and hence has at least two binary ones. The lower bound is then at least $2e > e$. □

7 Specialization to A396844

Substitute $m = 4$ in Theorems 4.1, 5.1, and 6.1, and in Corollary 6.3. In the residue formula, write $j = 2k$ or $j = 2k + 1$. Then $2^j = 4^k$ or $2 \cdot 4^k$ respectively, and $(-1)^j = 1$ or -1. This proves precisely the two exceptional classes stated at the start, with every other index handled by the second case of Theorem 5.1. The powers of two are strictly increasing, so these exceptional classes are disjoint.

8 Formal proof and scope

The accompanying entry file is `lean/Entries/A396844.lean`. It states the results with explicit Lean types and supplies proof terms using the complete shared implementation included in the collection. The manuscript is self-contained mathematically; the Lean entry file uses those ordinary library imports rather than duplicating the full foundation. The complete original acceptance layer is retained in `lean/Hanna/Acceptance.lean`. The shared endpoints `Hanna.Acceptance.inverse_binary_divisibility_integer` and `Hanna.Acceptance.minimum_prime_exponent_iff` give the general inverse-divisibility and minimum-exponent assertions used in this paper. The counterexample at $n = 17$ follows by taking $k = 2$ in the checked `residues_parameter_four` classification.

The entry's named theorem interfaces are listed below. The namespace of each is `FinalCollection.A396844`.

- `source_existence_uniqueness`
- `even_positive_tail`
- `binary_divisibility`
- `residues_parameter_four`

The preserved dependencies also contain all supporting constructions, coefficient identities, map-compatibility results, and inverse proofs. The collection pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The verification report records fresh compilation and the transitive axiom audit. Finite numerical checks corroborate the definitions and indexing; the universally quantified Lean theorems establish the all-index claims. No assertion of worldwide novelty, OEIS acceptance, publication, or completed human review follows from these checks. The prose may use elementary identities that the formal proof establishes by a different algebraic route.

References

[1] P. D. Hanna, *A265940*, The On-Line Encyclopedia of Integer Sequences, 19 December 2015. `https://oeis.org/A265940`. Accessed 22 October 2026.

[2] P. D. Hanna, *A271958*, The On-Line Encyclopedia of Integer Sequences, 17 April 2016. `https://oeis.org/A271958`. Accessed 22 October 2026.

[3] P. D. Hanna, *A396843*, The On-Line Encyclopedia of Integer Sequences, 1 July 2026. `https://oeis.org/A396843`. Accessed 22 October 2026.

[4] P. D. Hanna, *A396844*, The On-Line Encyclopedia of Integer Sequences, 3 July 2026. `https://oeis.org/A396844`. Accessed 22 October 2026.

[5] P. D. Hanna, *A396845*, The On-Line Encyclopedia of Integer Sequences, 3 July 2026. `https://oeis.org/A396845`. Accessed 22 October 2026.